



# Introduction to TensorFlow and Keras

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# Agenda

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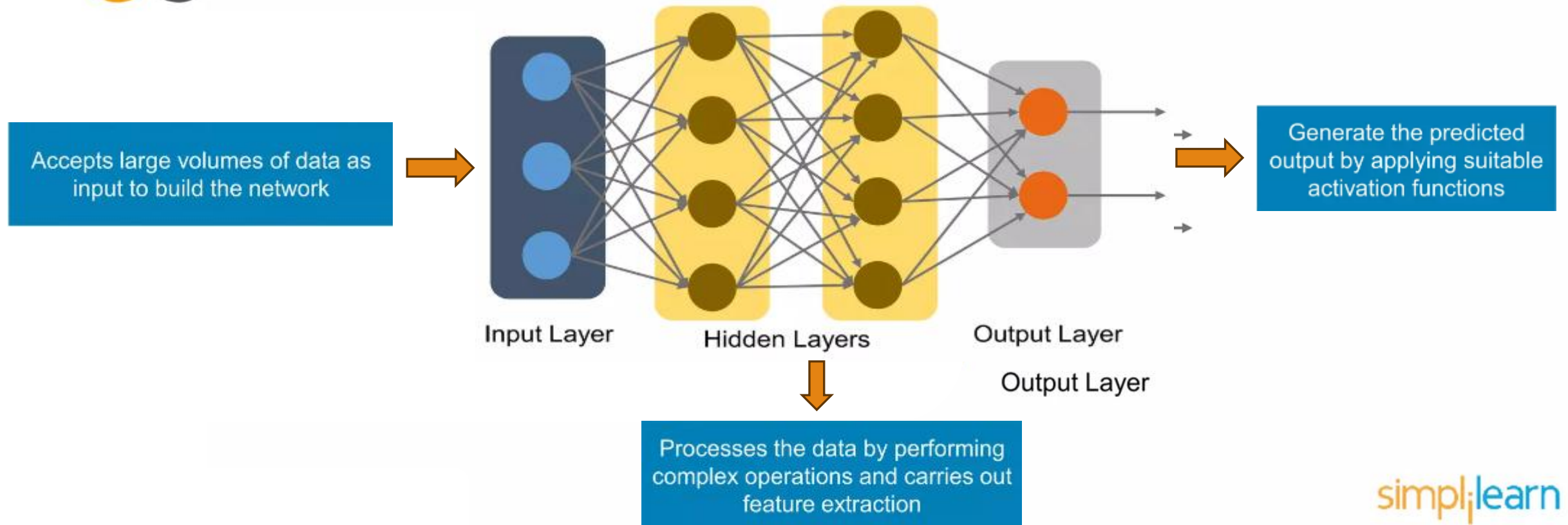
1. What is deep learning ?



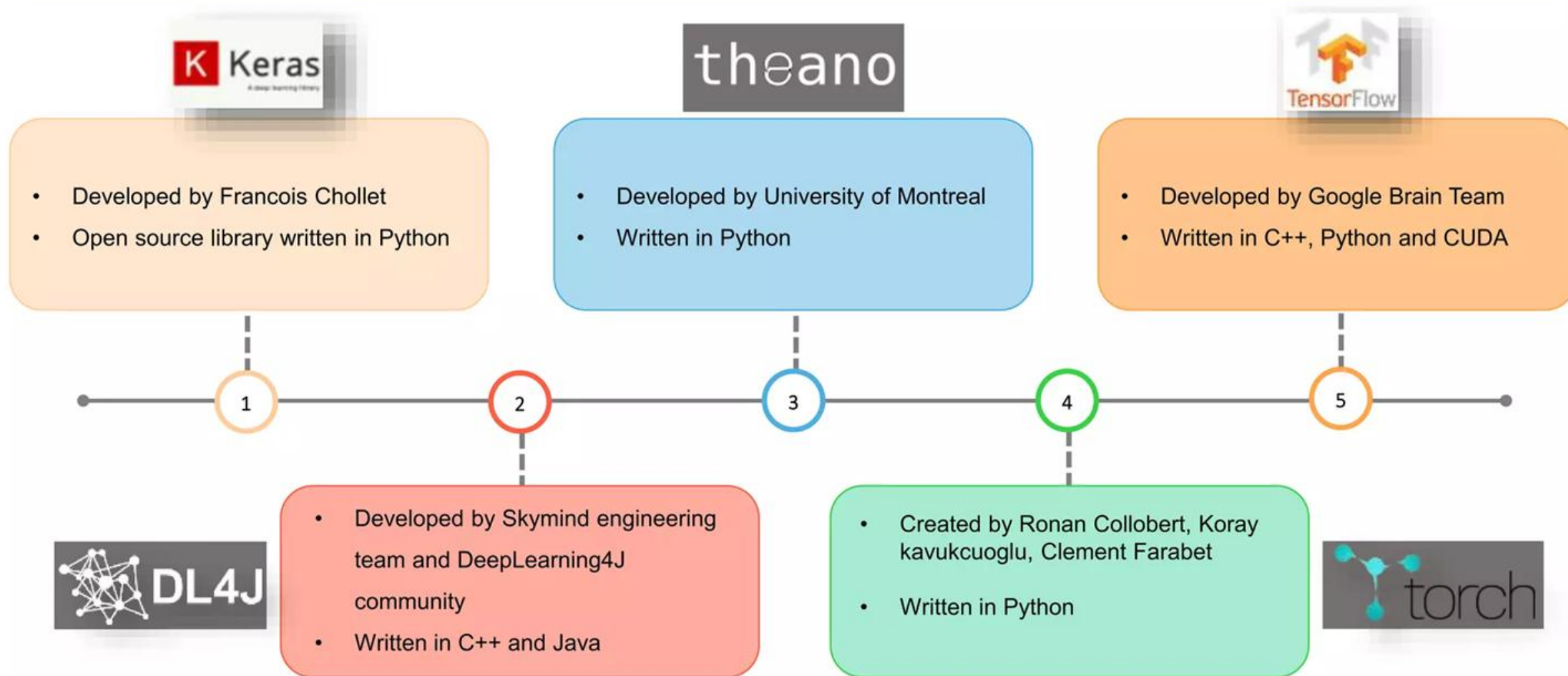
# What is Deep Learning



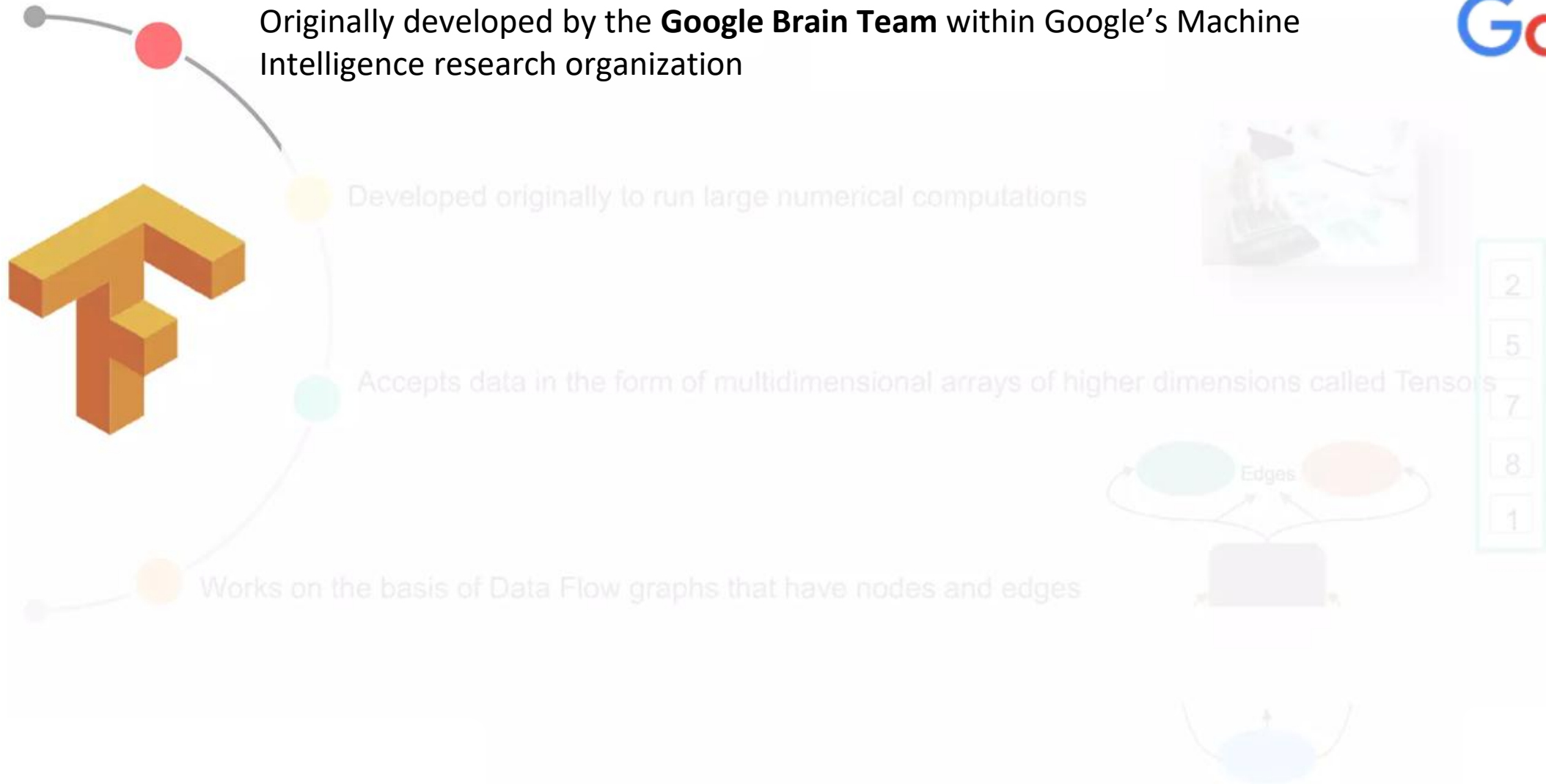
Deep Learning is a subset of Machine Learning and it works on the structure and functions of a human brain. It learns from data that is unstructured and uses complex algorithms to train a neural net.



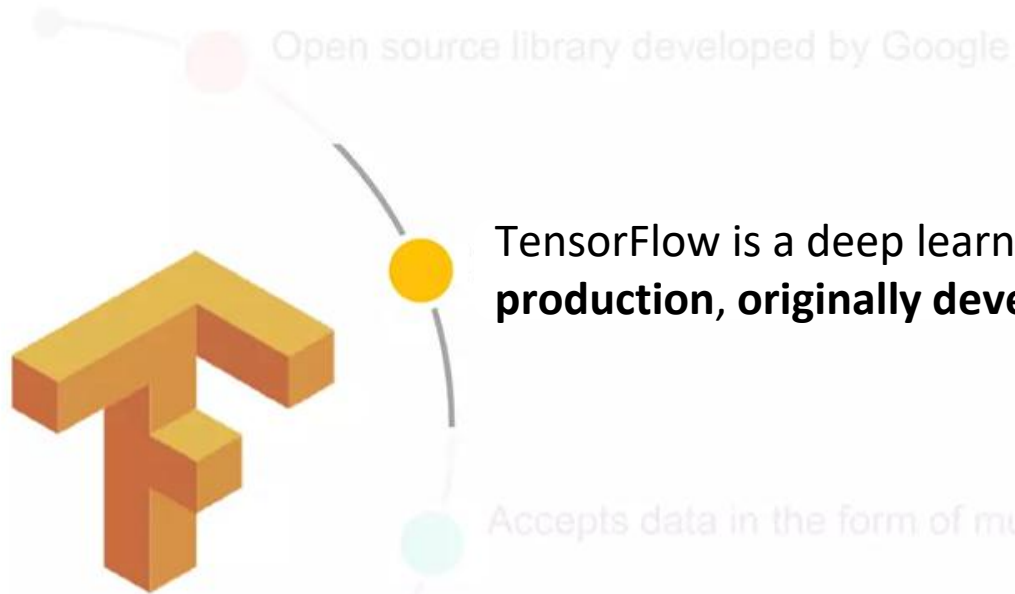
# Top Deep Learning Libraries



# What is TensorFlow?



# What is TensorFlow?



Open source library developed by Google



TensorFlow is a deep learning framework adopted both in **research** and **production**, originally developed to run large numerical computations

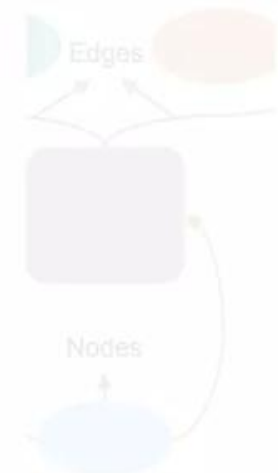


Accepts data in the form of multidimensional arrays of high

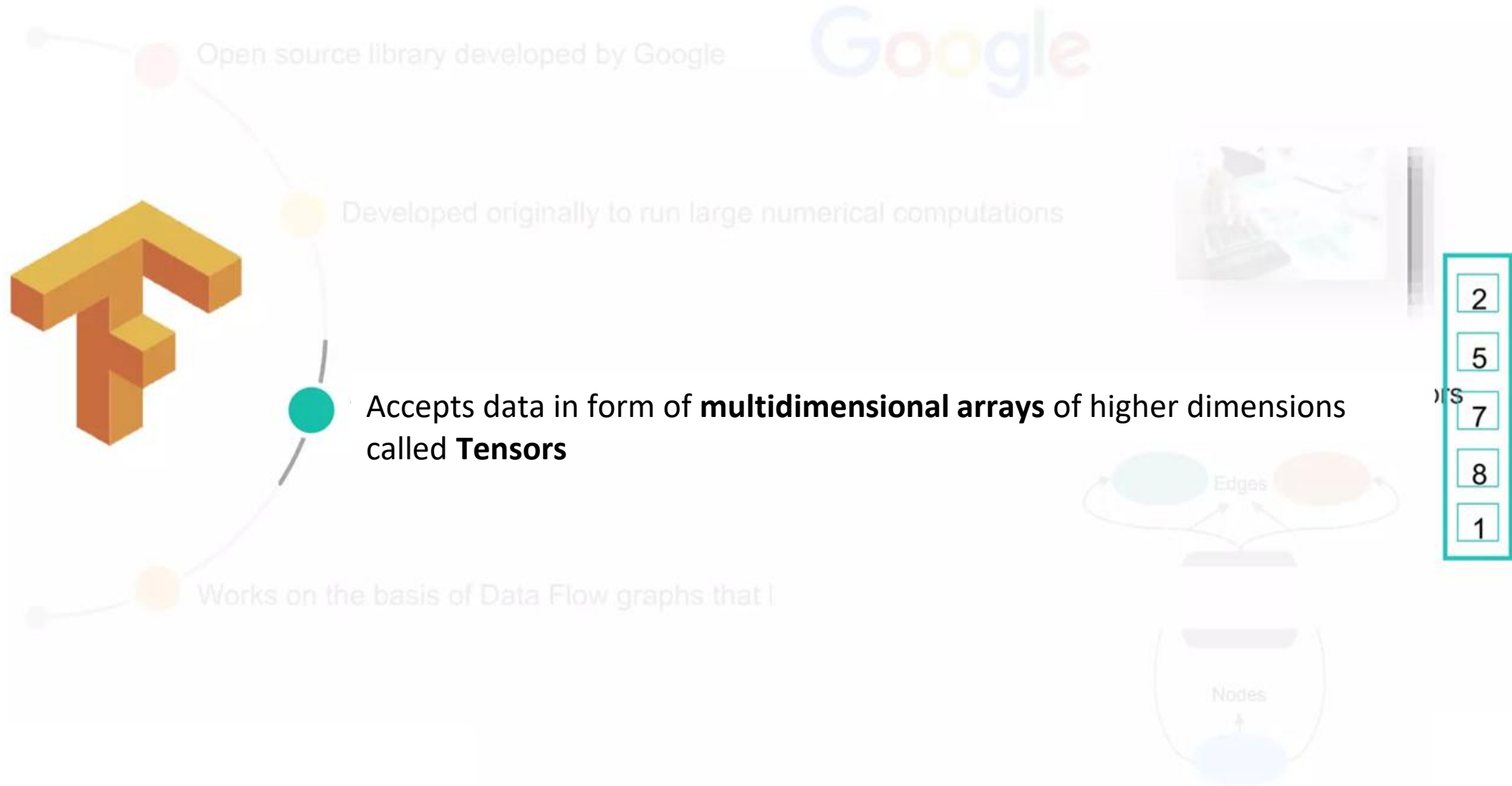
dimensions called tensors



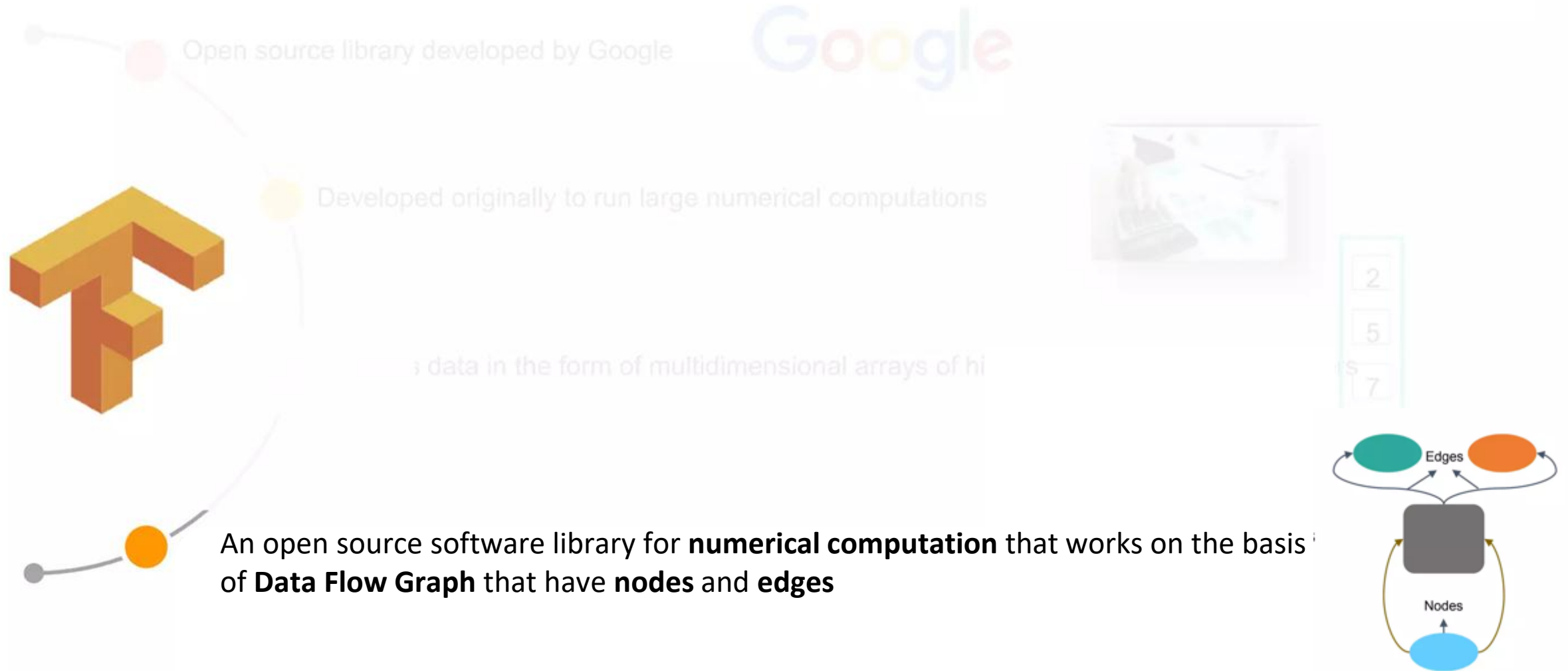
Work on the basis of Data Flow graphs that have nodes and edges



# What is TensorFlow?



# What is TensorFlow?



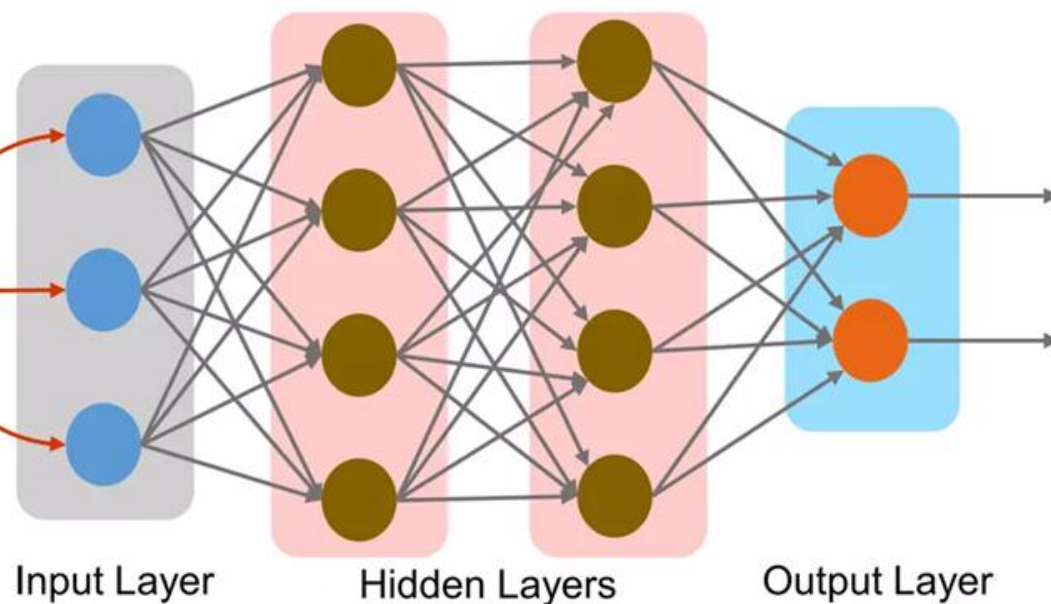
# What are Tensors?

## Tensors

*Tensor is a generalization of vectors and matrices of potentially higher dimensions. Arrays of data with different dimensions and ranks that are fed as input to the neural network are called Tensors.*

Data in the form of arrays is fed as input to the network

|   |   |   |
|---|---|---|
| 3 | 5 | 4 |
| 2 | 6 | 7 |
| 8 | 1 | 3 |
| 2 | 5 | 9 |
| 1 | 4 | 2 |

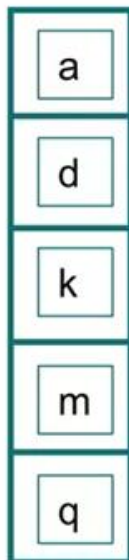


# What are Tensors?

Tensors

*Tensor is a generalization of vectors and matrices of potentially higher dimensions. Arrays of data with different dimensions and ranks that are fed as input to the neural network are called Tensors.*

Different Dimensions



Tensor of Dimensions[5]

# What are Tensors?

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Different Dimensions



|   |   |   |   |
|---|---|---|---|
| 1 | 3 | 4 | 7 |
| 9 | 7 | 3 | 2 |
| 8 | 4 | 1 | 6 |
| 6 | 3 | 9 | 1 |
| 3 | 1 | 5 | 9 |

Tensor of Dimensions[5,4]

# What are Tensors?

Tensors

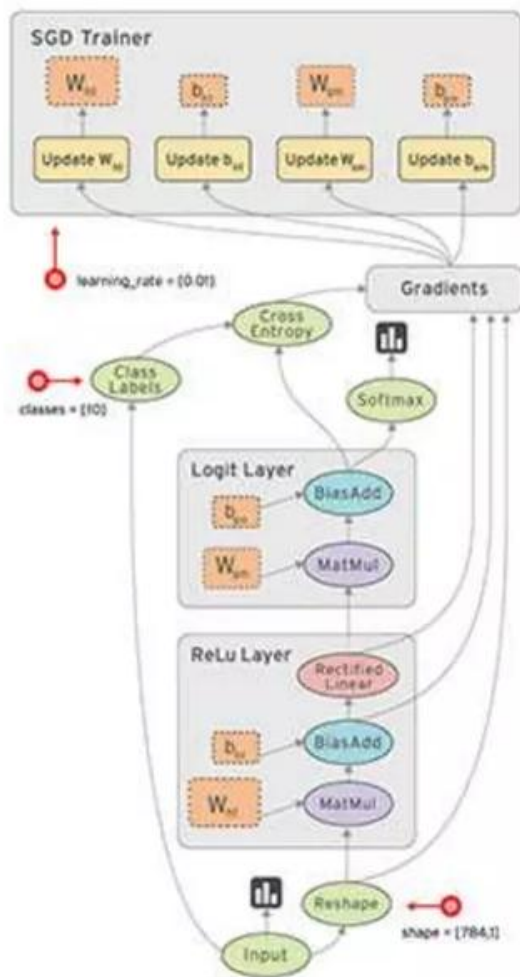
*Tensor is a generalization of vectors and matrices of potentially higher dimensions. Arrays of data with different dimensions and ranks that are fed as input to the neural network are called Tensors.*

Different Dimensions



Tensor of Dimensions[3,3,3]

# What is a Data Flow graph?



Each computation in TensorFlow is represented as a Data Flow Graph

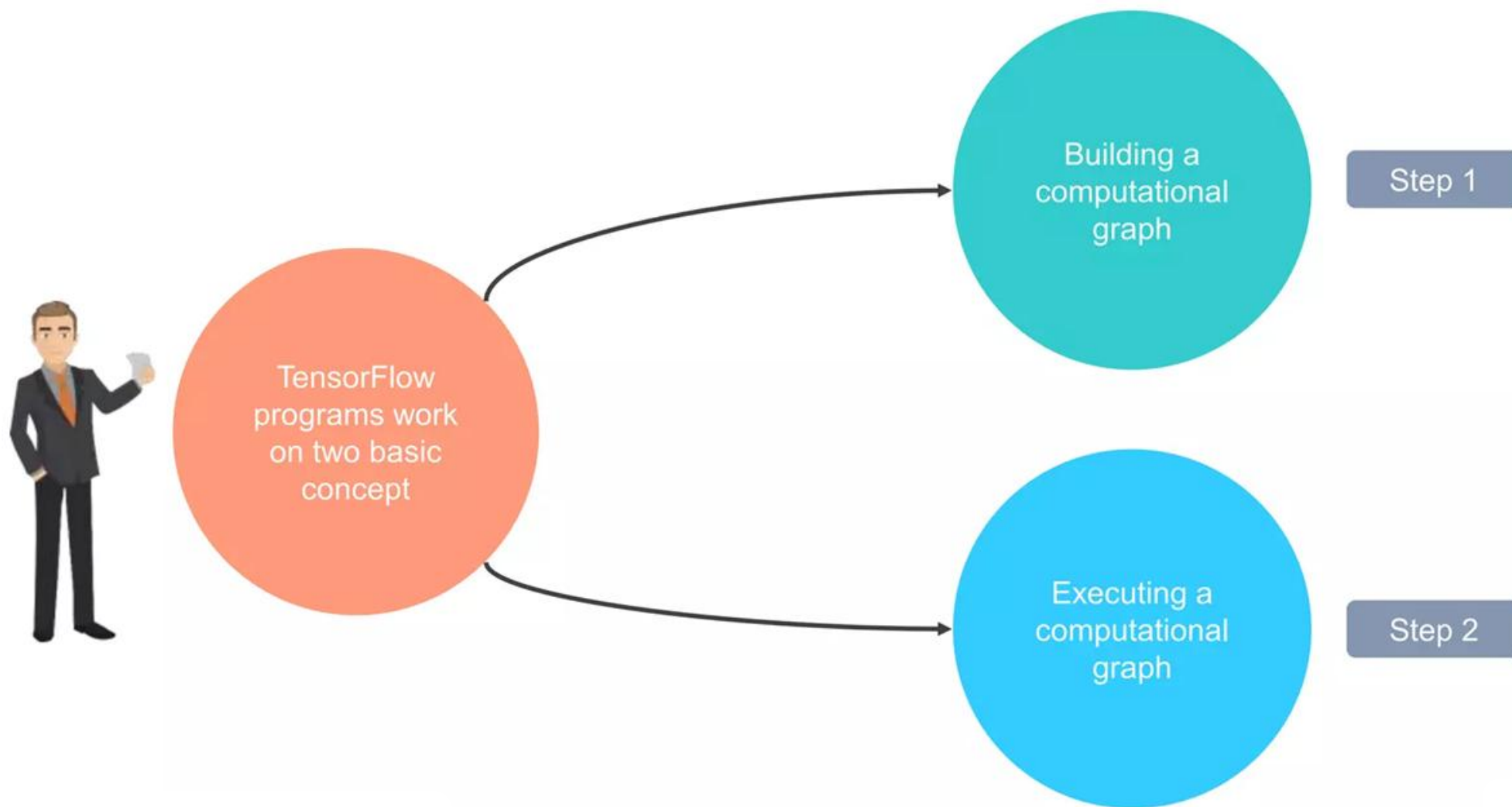
Each node in the graph represents a mathematical operation (add, subtract, multiply, etc) and each edge represents multidimensional arrays (Tensors)

Computational Graph is the graph of programming logic which Tensorflow builds in the memory

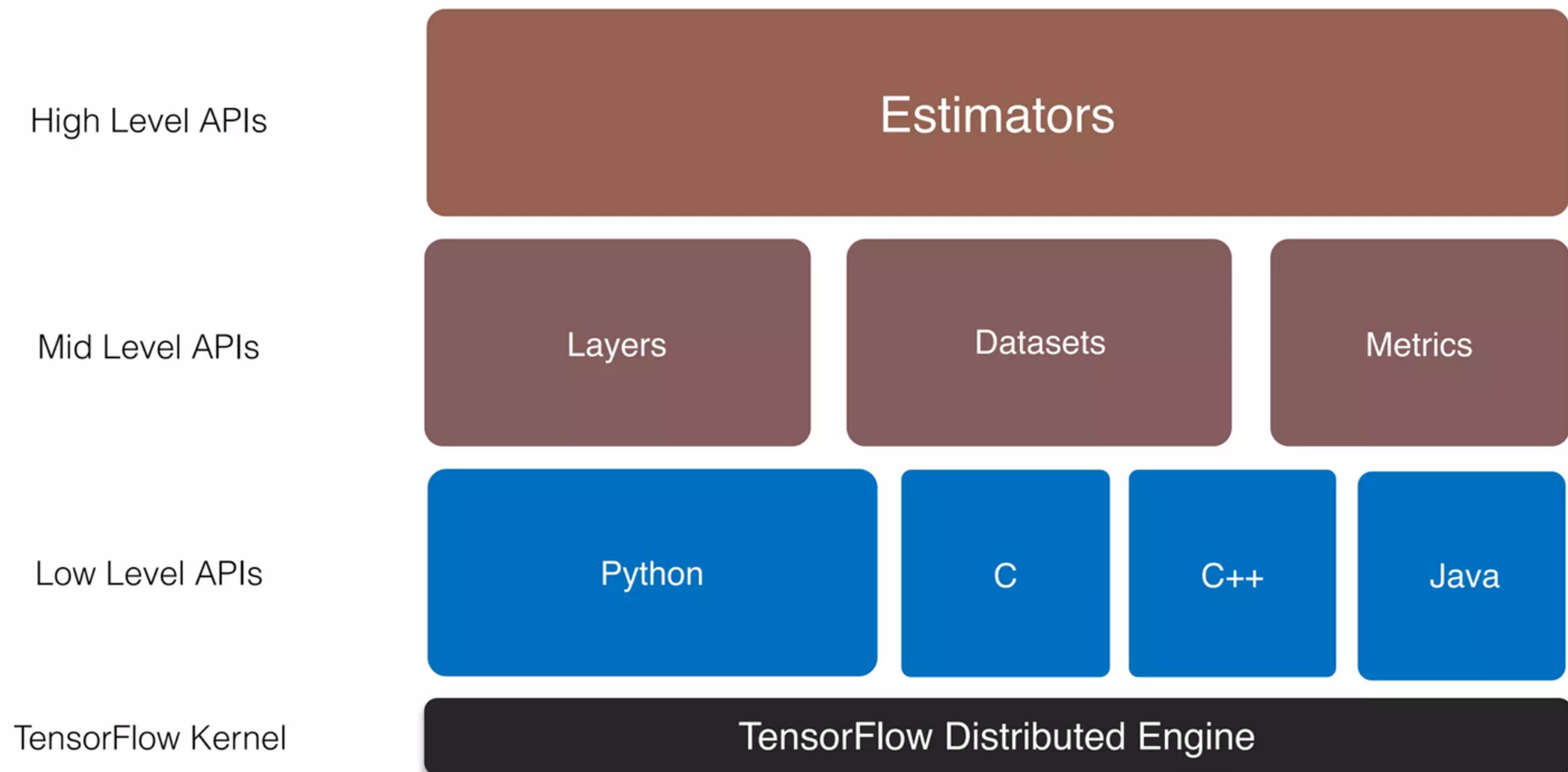
Enables creating large scale neural networks as computing can be distributed across several CPU's or GPU's

# Program Elements in TensorFlow

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# TensorFlow Architecture



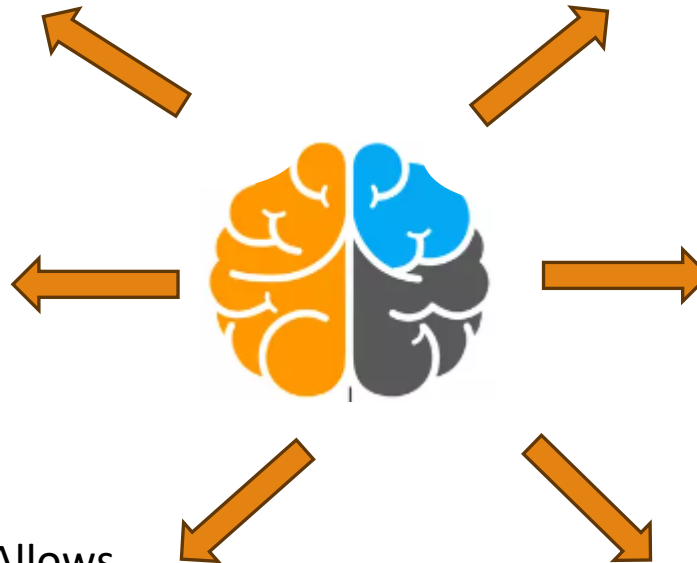
# Why TensorFlow?



It is open source and backed by a large community of software engineers including google

TensorFlow supports both CPU's and GPU's computing devices

Allows parallel neural networking. Allows batch normalization and optimization of algorithms



Provides both C++ and Python API's that makes it easier to work on

Has a faster compilation time than other Deep Learning Libraries like Torch,etc

Provides the visualization of graph which is not an option with Scikit or NumPy

# What is Keras ?

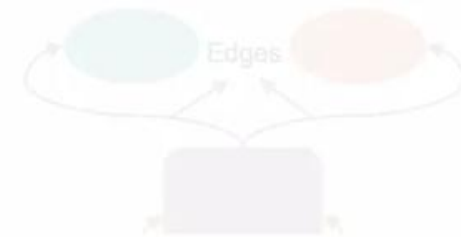
Keras is a High-level neural networks API, Written in Python



Developed originally to run large numerical computations

Accepts data in the form of multidimensional arrays of higher dimensions called Tensors

Works on the basis of Data Flow graphs that have nodes and edges



# What is Keras ?



Open source library developed by Google

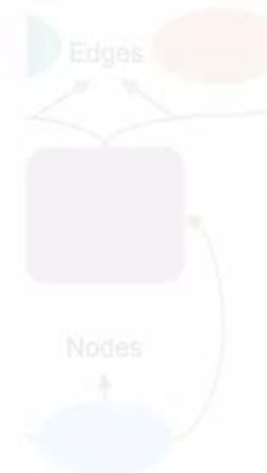


Keras minimizes the number of user actions required for common use cases, and provides clear and actionable feedback upon user error

Accepts data in the form of multidimensional arrays of high

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Work on the basis of Data Flow graphs that have nodes and edges



# What is Keras ?



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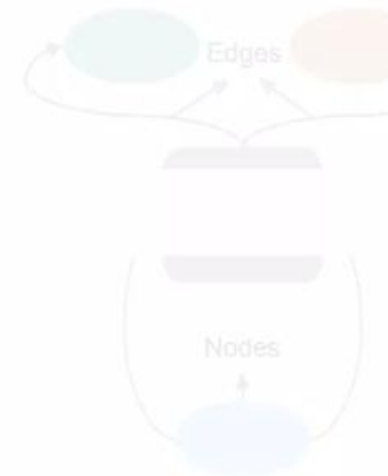


Developed originally to run large numerical computations



Allows for easy and fast prototyping (through user friendliness, modularity and extensibility)

Works on the basis of Data Flow graphs that



# What is Keras ?



Open source library developed by Google



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data in the form of multidimensional arrays of hi



Supports integration with TensorFlow, Theano and Runs seamlessly on CPU and GPUs

# Keras - Architecture

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Keras API spec

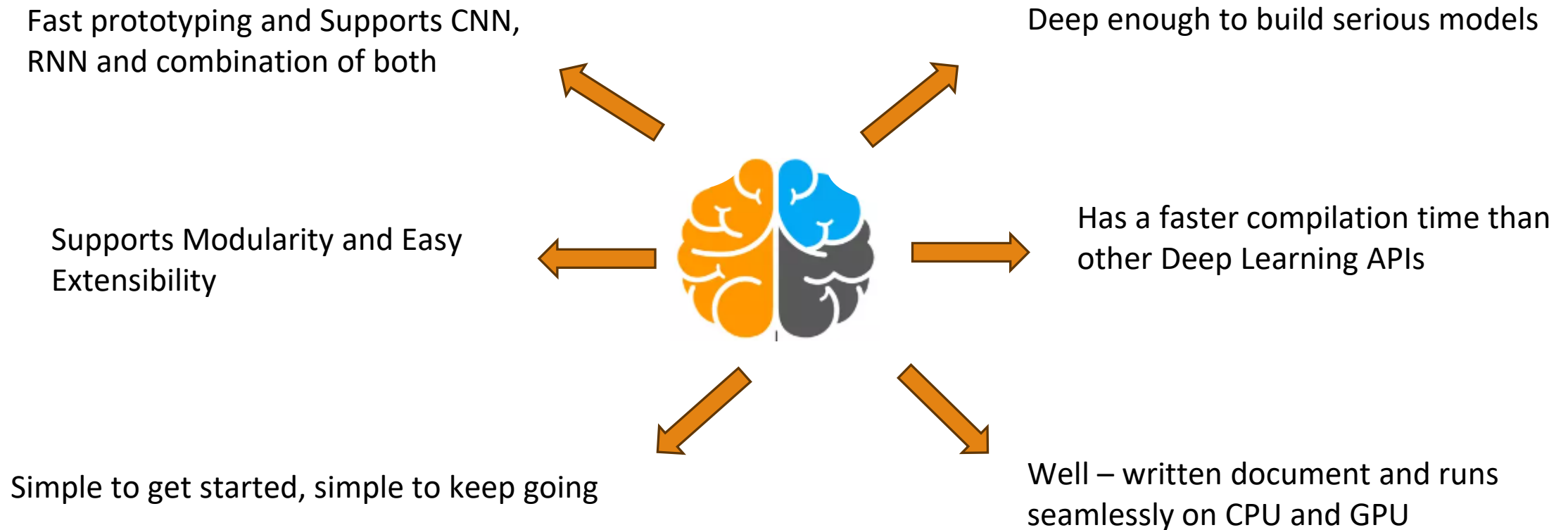
TensorFlow

Theano

CNTK

TensorFlow  
workflow

# Why Keras ?



THANK YOU !!